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SSAPy - Space Situational Awareness for Python

Joshua E. Meyers [1,2], **Michael D. Schneider** [3], **Julia T. Ebert** [4], **Edward F. Schlafly** [5], **Travis Yeager** [3], **Alexx Perloff** [3], **Daniel Merl** [3], **Noah Lifset** [6], **Jason Bernstein** [3], **William A. Dawson** [3], **Nathan Golovich** [3], **Denvir Higgins** [3], **Peter McGill** [3¶], **Caleb Miller** [3], **and Kerianne Pruett** [3]

1 SLAC National Accelerator Laboratory, 2575 Sand Hill Road, Menlo Park, CA 94025, USA **2** Kavli Institute for Particle Astrophysics and Cosmology, Stanford University, 452 Lomita Mall, Stanford, CA 94035, USA **3** Lawrence Livermore National Laboratory, 7000 East Ave., Livermore, CA 94550, USA **4** Fleet Robotics, 21 Properzi Way, Somerville, MA 02143, USA **5** Space Telescope Science Institute, 3700 San Martin Drive, Baltimore, MD 21218, USA **6** University of Texas at Austin, 2515 Speedway, Austin, TX 78712, USA ¶ Corresponding author

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Software

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Summary

SSAPy is a fast and flexible orbit modeling and analysis tool for orbits spanning from low-Earth into the cislunar regime. Orbits can be flexibly specified from common input formats such as Keplerian elements or two-line element data files. SSAPy allows users to model satellites and specify parameters such as satellite area, mass, and drag coefficients. SSAPy includes a customizable force propagation with a range of Earth, Lunar, radiation, atmospheric, and maneuvering models. SSAPy makes use of various community integration methods and can calculate time-evolved orbital quantities, including satellite magnitudes and state vectors. Users can specify various space- and ground-based observation models with support for multiple coordinate and reference frames. SSAPy also supports orbit analysis and propagation methods such as multiple hypothesis tracking and has built-in uncertainty quantification. The majority of SSAPy's methods are vectorized and parallelizable, allowing effective use of high performance computer (HPC) systems. Finally, SSAPy has plotting functionality, allowing users to visualize orbits and trajectories, an example of which is shown in Figures 1 and 2.

SSAPy has been used for the classification of cislunar (Higgins et al., 2024), and closely-spaced (Pruett et al., 2024), orbits as well as for studying the long-term stability of orbits in cislunar space (Yeager et al., 2023). SSAPy has also been used to build a case study for rare events analysis in the context of satellites passing close to each other in space (Bernstein et al., 2021; Miller et al., 2022).

Statement of need

Cislunar space is a region between earth out to beyond the Moon's orbit that includes the Lagrange points. This region of space is of growing importance to scientific and other space exploration endeavors (e.g., Duggan et al., 2019). Understanding, mapping, and modeling orbits through cislunar space is critical to all of these endeavors. The challenge for cislunar orbits is that n-body dynamics (e.g., gravitational forces from the Sun, Earth, Moon and other planets) are significant, leading to unpredictable and chaotic orbital motion. In this chaotic regime, orbits cannot be reduced to simple parametric descriptions making scalable orbit simulation and modeling a critical analysis tool (Yeager et al., 2023). Current orbit modeling software tools are predominantly used via graphical user interfaces (e.g., The General Mission Analysis Tool; Hughes et al. (2014) or the Systems Tool Kit) and are not optimized

for large scale simulation on HPC systems. Orbital modeling codes that can be run on HPC systems (e.g., REBOUND; Rein & Liu (2012)) lack full observable generation and modeling capabilities with uncertainty quantification. SSAPy, with its full-featured modeling framework and scalable, parallelizable functionality, fills the gap in the orbital software landscape.

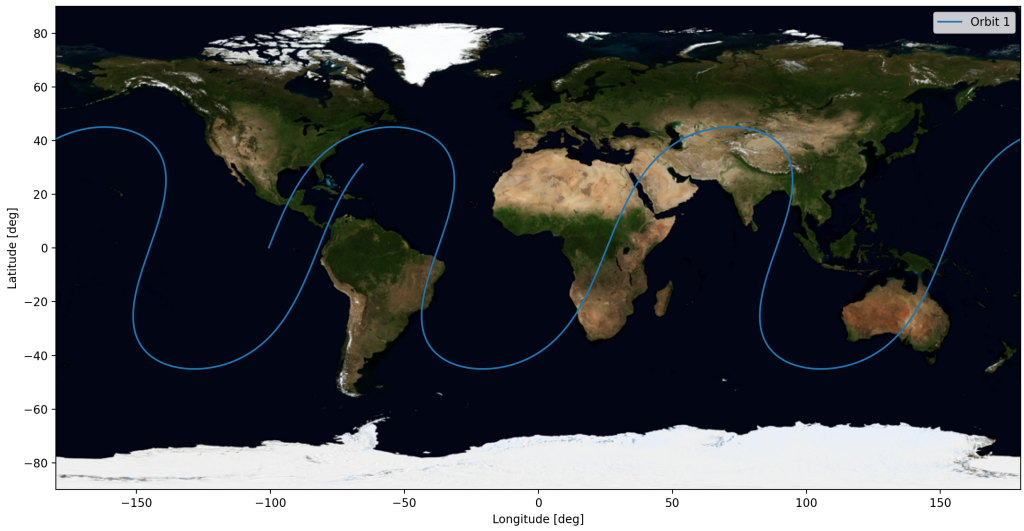

Figure 1: Example SSAPy visualization plot of an orbit ground track over the surface of the Earth. The 48 hour orbit has a semi-major axis of 0.75 GEO (27,000km), an eccentricity 0.2 and an inclination of 45 degrees.

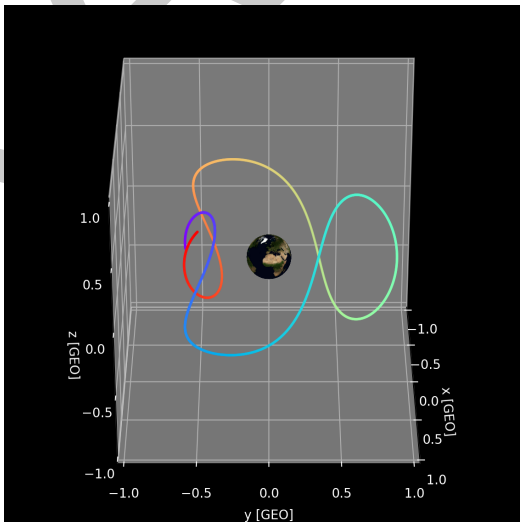

Figure 2: Example SSAPy visualization plot. Simulated trajectory of the orbit shown in Figure 1. The color on this plot represents time.

Acknowledgements

SSAPy depends on NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), matplotlib (Hunter, 2007), emcee (Foreman-Mackey et al., 2013), astropy (Astropy Collaboration et al., 2022), pyerfa (Kerkwijk et al., 2023), lmfit (Newville et al., 2024), and sqp4 (Vallado et al., 2006). We would like to thank Robert Armstrong and Iméne Goumiri for valuable contributions to this project. This work was performed under the auspices of the U.S. Department of Energy

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